Grundahl GX-125 Centrifugal Feed Pump

Operation and Maintenance Manual · Asset P-207 · Vestbjerg Processing Plant · Rev 4, March 2024

1. Equipment identification

Asset tag	P-207
Description	Feed pump, Line 2 inlet
Manufacturer	Grundahl Pumpeteknik A/S
Model	GX-125 end-suction centrifugal
Serial number	GX125-2019-04417
Year installed	2019
Location	Pump House 2, bay B, floor level
Functional location	VB-L2-PH2-207
Duty	Transfers process liquid from balance tank T-14 to Line 2 inlet manifold

2. Technical data

Parameter	Value
Rated flow	42 m³/h
Rated head	34 m
Motor	11 kW, 2900 rpm, 400 V 3-phase
Impeller diameter	178 mm, closed, bronze
Shaft seal	Mechanical seal, single, balanced — type GX-MS-32
Bearings	Drive end 6308-2RS; non-drive end 6206-2RS
Casing drain	G 3/8 plugged, bay B floor gully
Maximum working pressure	10 bar
Maximum liquid temperature	70 °C

3. Normal operation

P-207 runs continuously while Line 2 is in production. Start and stop are controlled from the Line 2 panel; local control is available at the pump via the isolator-mounted selector but should only be used during maintenance.

Suction is taken from balance tank T-14 through a 100 mm line fitted with a basket strainer (S-207) immediately upstream of the pump. Discharge passes through non-return valve NRV-207 to the Line 2 inlet manifold.

Normal running indications

- Discharge pressure 3.2 to 3.8 bar at rated flow
- Motor current 17 to 20 A
- Bearing temperature below 70 °C
- No visible leakage at the seal housing; a trace of moisture is acceptable, a drip is not

4. Routine maintenance schedule

Interval	Task
Weekly	Visual check for leakage, noise and vibration. Record discharge pressure
Monthly	Check and clean suction strainer S-207. Record differential pressure across strainer
6-monthly	Full service: bearing inspection, coupling alignment check, seal inspection, vibration reading
Annually	Impeller clearance check. Casing internal inspection
As required	Mechanical seal replacement

The 6-monthly service is the work order most commonly raised against this asset. Strainer cleaning is nominally a production task but has historically been carried out by maintenance.

5. Mechanical seal — type GX-MS-32

P-207 is fitted with a single balanced mechanical seal, part number GX-MS-32. The seal is a cartridge type and is replaced as a complete unit; individual faces are not serviced on site.

Part number	GX-MS-32
Description	Cartridge mechanical seal, single, balanced, 32 mm shaft
Faces	Silicon carbide / carbon
Elastomers	FKM
Expected service life	Approximately 3 years under rated duty
Stock	Held at Vestbjerg central store, bin 14-C. Normally two held

5.1 Indications of seal failure

- Continuous drip or spray from the seal housing drain
- Product visible on the baseplate below the pump
- Squealing or chattering at start-up
- Rapid rise in seal face temperature

5.2 Common causes of premature seal failure

Where a seal fails well short of its expected service life, the cause is rarely the seal itself. Check the following before fitting a replacement:

1. Dry running, including momentary loss of suction
2. Cavitation caused by restricted suction — a partially blocked suction strainer is the most common source
3. Shaft misalignment or excessive coupling wear
4. Solids in the pumped liquid
5. Incorrect fitting, particularly over-compression of the cartridge

Fitting a new seal without establishing the cause will usually result in a repeat failure.

6. Seal replacement procedure

Carry out only under an approved work order and after isolation in accordance with VB-SP-014. Estimated duration two hours with the pump already isolated and drained.

1. Confirm isolation and drain-down are complete and the permit is in place.
2. Disconnect the coupling guard and coupling spacer.

3. Remove the four M12 bolts securing the bearing housing to the casing.
4. Withdraw the back pull-out assembly. The casing remains on the pipework.
5. Release the cartridge seal setting clips and withdraw the seal from the shaft.
6. Inspect the shaft sleeve for scoring. Replace if scored.
7. Fit the replacement cartridge dry — do not grease the faces. Hand tighten the gland nuts evenly, then release the setting clips.
8. Refit the assembly, check coupling alignment with a dial gauge, and refit the guard.
9. Restore isolation, vent, and run for fifteen minutes observing the seal housing.
10. Record the work against the asset, including the suspected cause of the failure.

7. Suction strainer S-207

The basket strainer upstream of P-207 is a 100 mm Y-type with a stainless basket, 3 mm perforation. Differential pressure across the strainer should not exceed 0.3 bar.

The basket is removed by isolating at V-207A, venting, and releasing the six-bolt cover. The basket lifts out. Fibrous material tends to mat across the lower third of the basket and is not always visible from the inspection port.

A restricted strainer will not necessarily trip the pump. The first symptom is usually a change in the sound of the pump, followed over time by seal damage.

8. Spares

Item	Part number	Store location
Mechanical seal cartridge	GX-MS-32	14-C
Shaft sleeve	GX-SL-32	14-C
Casing gasket set	GX-GK-125	14-D
Bearing, drive end	6308-2RS	09-A
Bearing, non-drive end	6206-2RS	09-A
Strainer basket, S-207	ST-100-3MM	14-F